

Riccardo Barbieri

AI&ML Engineer



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rbarb



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Technical Skills

ML NLP MLOps Deep Learning

Data Pipelines LLM Workflows RAG

Experimentation Evaluation

Programming: Python, SQL, MATLAB, R

ML & DL: Scikit-learn, PyTorch, spaCy, Transformers

LLMs & Evals: LangChain, LangGraph, MLFlow, LangSmith, Vector DBs

Data Pipelines: Pandas, NumPy

Cloud & DevOps: AWS, Azure, Docker, Git

Transferable Skills

Project Management:

I manage complex projects involving ML, statistical modeling and AI. I collaborate with multidisciplinary teams, adapting to changing contexts to achieve ambitious goals, even with limited resources.

Creative Thinking and Problem Solving:

I develop innovative solutions for LLM-based systems such as RAG and agents, ML applications, predictive models and data pipelines. I combine an analytical approach with an experimental mindset, with a strong focus on scalability and generalizability.

Interpersonal Skills:

I collaborate effectively with research, data & analytics, data science, and AI engineering teams. I support junior colleagues and facilitate knowledge transfer. I am regarded as a positive and optimistic team member who helps build constructive working relationships.

Education

Doctoral Degree: Charité - Universitätsmedizin Berlin

Master in Cognitive Neuroscience: Università di Padova

Bachelor in Psychology: Università di Bologna

BitBang (2021 - Ongoing) ————— Bologna

AI Engineer (2024 - Ongoing)

Development, design, and deployment of systems based on LLM and ML.

- **Enterprise LLM workflows:** Technical lead in the development of RAG and agents serving multiple business units. Developed ingestion and retrieval pipelines with disambiguation flows and metadata-driven query resolution. Permission-aware retrieval to enforce access rules. Data anonymization through automatic entity extraction with lightweight local NLP models. Coordinated a cross-functional team to support the progressive evolution of the system into an enterprise AI platform. Stakeholder management across three different products.
- **Automation & classification:** Responsible for the development of a purchase-request classification system. Delivered a hybrid model combining random forest with vector retrieval module over procurement guidelines. Mentoring and progressive delegation of activities to a junior colleague.
- **Cash flow forecast:** Re-engineered a forecasting system that had grown into an unmaintainable collection of over 20 heterogeneous models spread across multiple data sources. The resulting system was more reproducible, maintainable and explainable. Outputs support weekly reporting to the finance department.

Data Scientist (2021 - 2024)

Implementation of data-driven solutions based on statistical and ML models.

- **Customer analytics & demand forecasting:** Contributed to the design of a forecasting model for raw material requirements based on historical sales and a multi-feature product matrix. Performed clustering analyses on e-commerce navigation data to identify traffic patterns across user segments, supporting marketing and personalization strategies.
- **Marketing performance measurement:** Developed an internal Marketing Mix Modeling framework supporting what-if simulations and budget optimization. Results were served through an Azure web application backed by a deployed model on Azure ML. The tool became part of the company's offering.
- **NLP & Deep Learning:** Developed a graph attention network for aspect-based opinion extraction in PyTorch Geometric, pre-trained on Wikipedia with a masked node prediction objective and fine-tuned on SemEval datasets, outperforming a rule-based dependency parsing baseline. Conducted topic modeling on survey data using sentence transformers.

Charité / BCCN (2016 - 2021) ————— Berlin

Researcher in Cognitive Neuroscience (2017 - 2021)

Experimental research on brain signals related to decision-making.

- **Experimental design & data analyses:** Designed and conducted behavioral and neuroimaging experiments on human participants. Developed data analysis pipelines using statistical and ML models.
- **Results dissemination:** Authored peer-reviewed publications, presented research at international conferences. Published anonymized datasets and reproducible code for the research community.

Research Intern (Feb 2016 - Sep 2016)

Computational neuroscience research conducted for master's thesis abroad.

- **Neuroimaging data analysis:** Experimentation with ML techniques and development of a classification pipeline starting from an existing dataset.
- **Lab work & programming:** MATLAB programming and collaboration within an interdisciplinary research team.